Linear Programing Notes

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1 Foundations of Linear Programming: Linear Algebra and Geometry

Linear Programming is fundamentally built upon Linear Algebra and Convex Geometry. In this section, we develop all the necessary mathematical foundations.

1.1 Vectors and Linear Systems

1.1.1 Vectors in $\mathbb{R}^n$

A vector in $\mathbb{R}^n$ is an ordered n-tuple:

$$x = (x_1, x_2, \ldots, x_n)$$

The standard inner product is defined as:

$$x^T y = \sum_{i=1}^{n} x_i y_i$$

Key Idea: Linear programming objective functions are linear functionals of the form $c^T x$.

1.1.2 Linear Independence

Vectors $v_1, \ldots, v_k \in \mathbb{R}^n$ are linearly independent if:

$$\sum_{i=1}^{k} \lambda_i v_i = 0 \implies \lambda_i = 0 \ \forall i.$$

Important: In LP, basic variables correspond to linearly independent columns of the constraint matrix.

1.1.3 Rank of a Matrix

Let $A \in \mathbb{R}^{m \times n}$. The rank of A is the dimension of its column space.

$$\text{rank}(A) = \text{maximum number of linearly independent columns of } A.$$

Key Result:

$$\text{rank}(A) \leq \min(m, n)$$

If $\text{rank}(A) = m$, then the constraints are linearly independent.

1.1.4 System of Linear Equations

Consider:

$$Ax = b$$

Consistency Condition: The system has a solution if and only if:

$$\text{rank}(A) = \text{rank}([A|b]).$$

If A has full row rank m, then solution set has dimension $n - m$.

1.2 Hyperplanes and Halfspaces

1.2.1 Hyperplane

A hyperplane in $\mathbb{R}^n$ is defined by:

$$a^T x = b$$

where $a \neq 0$.

Geometric Meaning: A hyperplane is an $(n-1)$-dimensional flat surface.
Important: Each constraint in LP defines a hyperplane.

1.2.2 Halfspace

A halfspace is defined by:

$$a^T x \leq b$$

It consists of all points lying on one side of the hyperplane.
Key Fact: Every halfspace is a convex set.

1.3 Convex Sets

1.3.1 Definition of Convex Set

A set $C \subseteq \mathbb{R}^n$ is convex if:

$$x, y \in C \Rightarrow \lambda x + (1 - \lambda)y \in C$$

for all $\lambda \in [0, 1]$.

Interpretation: The line segment between any two points lies entirely in the set.

1.3.2 Convex Combination

A convex combination of points $x_1, \ldots, x_k$ is:

$$\sum_{i=1}^{k} \lambda_i x_i$$

where:

$$\lambda_i \geq 0, \quad \sum \lambda_i = 1.$$

1.3.3 Convex Hull

The convex hull of a set S is:

$$\text{conv}(S) = \{\text{all convex combinations of points in } S\}.$$

Important: Every feasible region of an LP is convex.

1.4 Polyhedra and Polytopes

1.4.1 Polyhedron

A polyhedron is the intersection of finitely many halfspaces:

$$P = \{x \in \mathbb{R}^n : Ax \leq b\}.$$

Key Result: Every polyhedron is convex.

1.4.2 Polytope

A polytope is a bounded polyhedron.

Example: A simplex in $\mathbb{R}^n$.

1.5 Extreme Points and Faces

1.5.1 Extreme Point

Let C be convex. A point $x \in C$ is an extreme point if:

$$x = \lambda y + (1 - \lambda)z \Rightarrow y = z = x.$$

Interpretation: Cannot be written as a non-trivial convex combination.

1.5.2 Face of a Convex Set

A face F of a convex set C satisfies: If a segment in C has an interior point in F, then the whole segment lies in F.

Vertices are zero-dimensional faces.

1.6 Separation Theorem

Theorem (Hyperplane Separation):

If C is convex and $x \notin C$, then there exists a hyperplane separating x and C.

Importance in LP: Duality theory is based on separation of convex sets.

1.7 Dimension of a Polyhedron

If $P = \{x : Ax = b, x \geq 0\}$ and $\text{rank}(A) = m$, then:

$$\dim(P) = n - m.$$

Key Insight: The number of basic variables equals the rank of the constraint matrix.

Summary of Fundamental Results

- Every LP feasible region is a convex polyhedron.
- Extreme points correspond to intersections of linearly independent constraints.
- Rank determines degrees of freedom.
- Hyperplanes define constraints.
- Convexity guarantees global optimality of local optima.

Foundation Principle: Linear Programming = Linear Algebra + Convex Geometry.

2 Formulation of Linear Programming and Standard Forms

In this section, we formally define a Linear Programming (LP) problem and develop its various equivalent representations.

2.1 General Linear Programming Problem

A Linear Programming problem consists of:

- A linear objective function,
- Linear constraints,
- Non-negativity (or sign) restrictions.

General Form:

$$\text{Maximize } c^T x$$

subject to

$$Ax \leq b$$

$$x \geq 0$$

where:

- $x \in \mathbb{R}^n$ is the decision vector,
- $A \in \mathbb{R}^{m \times n}$ is the constraint matrix,
- $b \in \mathbb{R}^m$,
- $c \in \mathbb{R}^n$.

Key Observation: The feasible region

$$P = \{x \in \mathbb{R}^n : Ax \leq b, x \geq 0\}$$

is a convex polyhedron.

2.2 Minimization Form

A minimization LP is written as:

$$\text{Minimize } c^T x$$

subject to

$$Ax \geq b, \quad x \geq 0.$$

Important: Any minimization problem can be converted into a maximization problem by multiplying the objective by -1.

2.3 Standard Equality Form

For theoretical development (especially simplex method), we use the **Standard Equality Form**:

$$\text{Maximize } c^T x$$

subject to

$$Ax = b,$$

$$x \geq 0.$$

Here A has full row rank m.
Why Equality Form?
Because:

- Basic feasible solutions are defined via equality systems.
- Extreme points arise from intersecting m linearly independent constraints.

2.4 Conversion to Standard Form

2.4.1 1. Inequality Constraints

If we have:

$$a_i^T x \leq b_i,$$

introduce a **slack variable** $s_i \geq 0$:

$$a_i^T x + s_i = b_i.$$

If:

$$a_i^T x \geq b_i,$$

introduce a **surplus variable** $s_i \geq 0$:

$$a_i^T x - s_i = b_i.$$

Key Idea: Slack variables measure unused resources.

2.5 Unrestricted Variables

If a variable x_j is unrestricted in sign:

$$x_j \in \mathbb{R},$$

we replace it by:

$$x_j = x_j^+ - x_j^-,$$

where

$$x_j^+, x_j^- \geq 0.$$

This transformation preserves linearity.

2.6 Artificial Variables

Sometimes, after introducing surplus variables, we cannot immediately obtain a feasible basis.

We then introduce **artificial variables**.

Example:

$$a_i^T x - s_i = b_i.$$

We add artificial variable $a_i \geq 0$:

$$a_i^T x - s_i + a_i = b_i.$$

Artificial variables are removed using:

- Big-M method
- Two-phase method

2.7 Feasible Region

The feasible set is:

$$P = \{x \in \mathbb{R}^n : Ax = b, x \geq 0\}.$$

Properties:

- P is convex.
- P is a polyhedron.
- If bounded, P is a polytope.

2.8 Canonical Form

Another commonly used representation is:

$$\text{Maximize } z = c^T x$$

subject to

$$Ax + s = b,$$

$$x, s \geq 0.$$

Here s represents slack variables.

2.9 Fundamental Structural Insight

Any LP can be written in the compact form:

$$\max\{c^T x : x \in P\},$$

where P is a convex polyhedron.

Central Idea: Linear programming is optimization of a linear functional over a convex polyhedron.

Important Notes

- Equality form is best for theoretical analysis.
- Slack variables convert geometry into algebra.
- Rank of A determines number of basic variables.
- LP structure depends entirely on linearity.
- Convexity ensures that local optimum is global optimum.

Core Structural Principle:
LP = Maximize a linear functional over a convex polyhedron.

3 Basic Solutions and Basic Feasible Solutions

In this section we develop the central algebraic structure of Linear Programming. Everything in the simplex method and LP geometry is built upon these concepts.

We consider the LP in standard equality form:

$$\max\{c^T x : Ax = b,\ x \geq 0\},$$

where $A \in \mathbb{R}^{m \times n}$ has rank m and $m \leq n$.

3.1 Basic Solutions

3.1.1 Definition (Basis)

Let $A = [A_1\ A_2\ \dots\ A_n]$ denote the columns of A.

A set of indices $B \subseteq \{1, \dots, n\}$ with $|B| = m$ is called a **basis** if the columns $\{A_j : j \in B\}$ are linearly independent.

Let A_B denote the $m \times m$ matrix formed by these columns.

Since rank$(A) = m$, such subsets always exist.

3.1.2 Definition (Basic Solution)

Given a basis B:

- Set $x_j = 0$ for all $j \notin B$ (nonbasic variables).
- Solve $A_B x_B = b$ for the basic variables.

If A_B is invertible, then:

$$x_B = A_B^{-1} b.$$

The vector x constructed in this way is called a **basic solution**.

Important: A basic solution satisfies $Ax = b$, but may not satisfy $x \geq 0$.

3.2 Basic Feasible Solutions (BFS)

3.2.1 Definition

A basic solution is called a **basic feasible solution (BFS)** if additionally:

$$x \geq 0.$$

Thus, a BFS is a solution obtained from a basis such that:

$$x_B = A_B^{-1} b \geq 0.$$

Key Insight: Every BFS corresponds to selecting m linearly independent columns of A.

3.3 Number of Basic Solutions

The total number of possible bases is:

$$\binom{n}{m}.$$

Not all bases yield feasible solutions.
Hence:

$$\text{Number of BFS} \leq \binom{n}{m}.$$

3.4 Degeneracy

3.4.1 Definition

A basic feasible solution is **degenerate** if one or more basic variables are zero.
That is,

$$x_B = A_B^{-1} b$$

has at least one zero component.

Geometric Meaning: More than m constraints are active at that point.

Important Consequences:

- Multiple bases may correspond to the same extreme point.
- Degeneracy may cause cycling in the simplex method.

3.5 Geometric Interpretation

Let

$$P = \{x \in \mathbb{R}^n : Ax = b, x \geq 0\}.$$

Then:

- A basic feasible solution corresponds to an intersection of m linearly independent constraints.
- Algebraically: choosing m independent columns.
- Geometrically: intersection of m hyperplanes.

3.6 Theorem (Equivalence Theorem)

Let $P = \{x : Ax = b, x \geq 0\}$ with $\text{rank}(A) = m$.
Then:

$$x \text{ is a BFS} \iff x \text{ is an extreme point of } P.$$

This is one of the most important theorems in LP.
It connects:

- Algebra (basis selection)
- Geometry (extreme points)
- Optimization (optimal solutions)

3.7 Proof Sketch of Equivalence Theorem

($\Rightarrow$)

If x is a BFS, suppose:

$$x = \lambda y + (1 - \lambda)z.$$

Since $Ax = b$ and columns in B are independent, it follows that $y = z = x$. Thus x is extreme.

($\Leftarrow$)

If x is extreme, the columns corresponding to its positive components must be linearly independent. Hence they form a basis.

Therefore x is a BFS.

3.8 Fundamental Theorem of Linear Programming

If the LP

$$\max\{c^T x : Ax = b, x \geq 0\}$$

has an optimal solution, then it has an optimal basic feasible solution.

Meaning: The search for optimal solutions can be restricted to BFS only.

Why?

Because:

- The feasible region is convex.
- A linear function attains its maximum at extreme points.
- Extreme points are exactly BFS.

Important Structural Summary

- Basis = selection of m linearly independent columns.
- Basic solution = set nonbasic variables to zero.
- BFS = basic solution + nonnegativity.
- Extreme points $\leftrightarrow$ BFS.
- Optimal solution (if exists) occurs at a BFS.

Core Structural Insight:

Linear Programming reduces infinite optimization to a finite search over basic feasible solutions.

4 Geometry of Polyhedra and Extreme Points

In this section, we study the geometric structure of the feasible region of a Linear Program. The feasible set is a convex polyhedron, and understanding its structure is essential for deep theoretical insight.

4.1 Polyhedra

4.1.1 Definition

A **polyhedron** in $\mathbb{R}^n$ is a set of the form:

$$P = \{x \in \mathbb{R}^n : Ax \leq b\},$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.
Important Properties:

- A polyhedron is an intersection of finitely many halfspaces.
- Every polyhedron is convex.
- A polyhedron may be bounded or unbounded.

4.1.2 Polytope

A **polytope** is a bounded polyhedron.
Example:

$$\{x : Ax \leq b \text{ and } x \geq 0 \text{ and bounded}\}$$

4.2 Affine Sets and Dimension

4.2.1 Affine Set

A set S is affine if:

$$x, y \in S \Rightarrow \lambda x + (1 - \lambda)y \in S \quad \forall \lambda \in \mathbb{R}.$$

Note: Affine combinations allow λ outside $[0, 1]$.

4.2.2 Affine Hull

The affine hull of a set S is the smallest affine set containing S.

4.2.3 Dimension of a Polyhedron

If

$$P = \{x : Ax = b, x \geq 0\}$$

and $\text{rank}(A) = m$, then:

$$\dim(P) = n - m.$$

Geometric Meaning: Each independent equality constraint reduces dimension by one.

4.3 Faces of a Polyhedron

4.3.1 Definition

Let P be a polyhedron. A subset $F \subseteq P$ is a **face** of P if there exists a vector c such that:

$$F = \{x \in P : c^T x = \max_{y \in P} c^T y\}.$$

Interpretation: A face consists of all maximizers of some linear functional.

Examples:

- The entire polyhedron
- A facet (dimension $n - 1$ face)
- An edge
- A vertex (0-dimensional face)

4.4 Extreme Points

4.4.1 Definition

A point $x \in P$ is an **extreme point** if:

$$x = \lambda y + (1 - \lambda)z, \quad y, z \in P,\ \lambda \in (0, 1)$$

implies

$$y = z = x.$$

Interpretation: An extreme point cannot be written as a nontrivial convex combination.

4.5 Characterization of Extreme Points

Theorem:
Let

$$P = \{x : Ax = b, x \geq 0\}.$$

Then x is an extreme point of P if and only if the columns of A corresponding to positive components of x are linearly independent.

This gives the geometric meaning of BFS.

4.6 Supporting Hyperplanes

4.6.1 Definition

A hyperplane

$$H = \{x : c^T x = \alpha\}$$

is called a **supporting hyperplane** of P if:

$$c^T x \leq \alpha \quad \forall x \in P$$

and equality holds for at least one point in P.

Geometric Insight: At an optimal solution of LP, the objective hyperplane supports the feasible region.

4.7 Minkowski–Weyl Theorem

Theorem (Structure of Polyhedra):
Every polyhedron can be written as:

$$P = \text{conv}(V) + \text{cone}(R),$$

where:

- V is the set of extreme points,
- R is the set of extreme rays.

Interpretation:
A polyhedron is generated by:

- Convex combinations of vertices,
- Nonnegative combinations of rays.

If P is bounded (polytope), then:

$$P = \text{conv}(V).$$

4.8 Carathéodory's Theorem

Theorem:
If $x \in \text{conv}(S) \subseteq \mathbb{R}^n$, then x can be expressed as a convex combination of at most $n+1$ points of S.

Importance in LP: Every feasible point can be expressed using finitely many extreme points.

4.9 Extreme Rays

If P is unbounded, there exist directions $d \neq 0$ such that:

$$x + \lambda d \in P \quad \forall \lambda \geq 0.$$

Such d are called **extreme rays**.

Connection to LP: If objective improves along an extreme ray, the LP is unbounded.

Geometric Summary of LP

- Feasible region = convex polyhedron.
- Vertices = basic feasible solutions.
- Optimal solutions occur at extreme points (if bounded).
- Unboundedness corresponds to extreme rays.
- Duality arises from supporting hyperplanes.

Deep Structural Insight:
Linear Programming is the study of linear functionals over convex polyhedra, whose structure is completely determined by extreme points and extreme rays.

5 The Simplex Method: Algebra and Geometry Combined

The simplex method is an algorithm that solves a linear program by moving from one basic feasible solution (BFS) to another, improving the objective value at each step.

We consider the standard form:

$$\max\{c^T x : Ax = b,\ x \geq 0\},$$

where $A \in \mathbb{R}^{m \times n}$ has rank m.

5.1 Basic and Nonbasic Variables

Let B be a basis (set of m indices). Let N denote the remaining indices.

Partition variables as:

$$x = (x_B, x_N).$$

Then:

$$A_B x_B + A_N x_N = b.$$

Since A_B is invertible:

$$x_B = A_B^{-1} b - A_B^{-1} A_N x_N.$$

Geometric Interpretation: Setting $x_N = 0$ gives the current BFS. Allowing x_N to increase moves along an edge of the polyhedron.

5.2 Objective Function Representation

Partition the cost vector:

$$c = (c_B, c_N).$$

Then:

$$c^T x = c_B^T x_B + c_N^T x_N.$$

Substitute for x_B:

$$c^T x = c_B^T A_B^{-1} b + \left(c_N^T - c_B^T A_B^{-1} A_N\right) x_N.$$

Define the **reduced costs**:

$$\bar{c}_N^T = c_N^T - c_B^T A_B^{-1} A_N.$$

Thus,

$$c^T x = c_B^T A_B^{-1} b + \bar{c}_N^T x_N.$$

Key Insight: Reduced costs determine whether moving along a nonbasic direction improves the objective.

5.3 Optimality Condition

Theorem (Optimality Criterion):
A BFS is optimal if and only if:

$$\bar{c}_j \leq 0 \quad \forall j \in N.$$

Reason:
If any $\bar{c}_j > 0$, increasing x_j increases the objective value.

Geometric Meaning:
The objective hyperplane supports the feasible region.

5.4 Entering and Leaving Variables

If some $\bar{c}_j > 0$, we select such j as the **entering variable**.
We increase x_j from zero.
To maintain feasibility:

$$x_B = A_B^{-1} b - A_B^{-1} A_j x_j \geq 0.$$

Let:

$$d = A_B^{-1} A_j.$$

Then feasibility requires:

$$x_B - d x_j \geq 0.$$

5.4.1 Ratio Test

To maintain $x_B \geq 0$:

$$x_j \leq \min_{i:d_i>0} \frac{(x_B)_i}{d_i}.$$

The basic variable achieving the minimum leaves the basis.

This defines the pivot step.

5.5 Unboundedness Criterion

If for some entering variable j:

$$d = A_B^{-1} A_j \leq 0,$$

then x_j can increase indefinitely.

Conclusion: The LP is unbounded.

5.6 Degeneracy and Cycling

If the minimum ratio is zero, the new BFS has the same objective value.
This is called **degeneracy**.

Degeneracy may cause:

- Stalling,
- Cycling.

Bland's Rule:
Choose the smallest indexed entering and leaving variables to prevent cycling.

5.7 Simplex Algorithm (Summary)

1. Start with an initial BFS.
2. Compute reduced costs.
3. If all reduced costs ≤ 0, stop (optimal).
4. Choose entering variable with $\bar{c}_j > 0$.
5. Perform ratio test.
6. Pivot (update basis).
7. Repeat.

5.8 Geometric Interpretation of Simplex

- Each BFS is a vertex of the polyhedron.
- Pivoting moves along an edge.
- Objective strictly increases (unless degenerate).
- Algorithm terminates at a supporting vertex.

Core Geometric Insight:
Simplex walks along edges of the polyhedron, moving from vertex to vertex, until it reaches one where no adjacent edge improves the objective.

5.9 Finite Termination

Since the number of BFS is finite (at most $\binom{n}{m}$), and objective strictly increases (except degeneracy), simplex terminates in finitely many steps (if cycling is avoided).

Important Structural Result:
Simplex reduces infinite optimization to finite combinatorial search over bases.

6 Duality Theory: The Deep Structural Core of Linear Programming

Duality is one of the most profound principles in optimization. Every linear program has an associated dual problem, and the interaction between primal and dual reveals the geometric structure of convex polyhedra.

6.1 The Primal Problem

Consider the primal LP in inequality form:

$$\text{(P)} \quad \max\{c^T x : Ax \leq b,\ x \geq 0\},$$

where

$$A \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m, \quad c \in \mathbb{R}^n.$$

6.2 The Dual Problem

The dual problem is defined as:

$$\text{(D)} \quad \min\{b^T y : A^T y \geq c,\ y \geq 0\}.$$

Structure Correspondence:

- Each primal constraint corresponds to a dual variable.
- Each primal variable corresponds to a dual constraint.
- "$\leq$" constraints in primal give $y \geq 0$.

6.3 Weak Duality Theorem

Theorem (Weak Duality):
If x is feasible for (P) and y is feasible for (D), then:

$$c^T x \leq b^T y.$$

Proof:
Since $Ax \leq b$ and $y \geq 0$,

$$y^T Ax \leq y^T b.$$

Since $A^T y \geq c$ and $x \geq 0$,

$$x^T A^T y \geq x^T c.$$

But:

$$x^T A^T y = y^T Ax.$$

Therefore,

$$c^T x \leq b^T y.$$

Interpretation:
Every dual feasible solution gives an upper bound for the primal maximization problem.

6.4 Strong Duality Theorem

Theorem (Strong Duality):
If either (P) or (D) has an optimal solution, then both have optimal solutions and:

$$\max(P) = \min(D).$$

This is the central theorem of LP.

Geometric Meaning:
At optimality, the objective hyperplane simultaneously supports the primal feasible region and the dual feasible region.

6.5 Complementary Slackness

Theorem (Complementary Slackness):
Let x^* and y^* be feasible solutions. They are optimal if and only if:

$$y_i^*(b_i - a_i^T x^*) = 0 \quad \forall i,$$

$$x_j^*((A^T y^*)_j - c_j) = 0 \quad \forall j.$$

Interpretation:

- Either a constraint is tight,
- Or its corresponding dual variable is zero.

This condition characterizes optimality.

6.6 Economic Interpretation

- Primal variables = activity levels.
- Dual variables = shadow prices.
- Dual feasibility ensures resources are priced consistently.
- Strong duality implies equilibrium pricing.

6.7 Farkas' Lemma

Theorem (Farkas' Lemma):
Exactly one of the following holds:

$$Ax = b,\ x \geq 0$$

or

$$y^T A \geq 0, \quad y^T b < 0.$$

Importance:
Farkas' Lemma is the algebraic foundation of duality. It is a theorem of alternatives.

6.8 Geometric Interpretation of Duality

Duality arises from hyperplane separation:

- The primal feasible region is convex.
- The optimal objective hyperplane supports it.
- The dual variables describe that supporting hyperplane.

Thus:

$$\text{Duality} = \text{Supporting Hyperplane Principle.}$$

6.9 Dual of Standard Equality Form

If primal is:

$$\max\{c^T x : Ax = b, x \geq 0\},$$

then dual is:

$$\min\{b^T y : A^T y \geq c\},$$

where y is unrestricted in sign.

Sign Rules Summary:

- Primal constraint $\leq \Rightarrow$ dual variable ≥ 0.
- Primal constraint $\geq \Rightarrow$ dual variable ≤ 0.
- Equality constraint $\Rightarrow$ dual variable free.
- Primal variable $\geq 0 \Rightarrow$ dual constraint $\geq$.
- Primal variable free $\Rightarrow$ dual constraint equality.

Deep Structural Summary

- Weak duality gives bounds.
- Strong duality gives equality.
- Complementary slackness gives optimality conditions.
- Farkas' Lemma is the foundation.
- Duality is geometric separation.

Ultimate Insight:
Linear Programming duality expresses the equivalence between:

Optimization of a linear functional
and
Existence of a supporting hyperplane.

7 Sensitivity Analysis and Post-Optimality Theory

After solving a Linear Programming problem, it is natural to ask how sensitive the optimal solution is to changes in the data.

Sensitivity analysis studies how variations in

$$A,\ b,\ c$$

affect optimality and feasibility.

7.1 Stability of the Optimal Basis

Let B be the optimal basis.

Recall:

$$x_B = A_B^{-1} b, \quad \bar{c}_N^T = c_N^T - c_B^T A_B^{-1} A_N.$$

Optimality requires:

$$x_B \geq 0, \quad \bar{c}_N \leq 0.$$

Thus:

- Feasibility depends on b,
- Optimality depends on reduced costs.

7.2 Change in Right-Hand Side Vector b

Suppose:

$$b \to b + \Delta b.$$

Then:

$$x_B = A_B^{-1}(b + \Delta b) = A_B^{-1} b + A_B^{-1} \Delta b.$$

The basis remains feasible if:

$$A_B^{-1}(b + \Delta b) \geq 0.$$

Important Insight:

As long as feasibility is preserved, the basis remains optimal (because reduced costs do not depend on b).

7.3 Shadow Prices

The dual solution corresponding to basis B is:

$$y^T = c_B^T A_B^{-1}.$$

Interpretation:
The vector y gives the rate of change of the optimal value with respect to b. Indeed:

$$z^* = c_B^T A_B^{-1} b = y^T b.$$

Thus,

$$\frac{\partial z^*}{\partial b_i} = y_i.$$

Hence: Dual variables are shadow prices.

7.4 Change in Objective Coefficients c

Suppose:

$$c \to c + \Delta c.$$

Reduced costs become:

$$\bar{c}_N^T = (c_N + \Delta c_N)^T - (c_B + \Delta c_B)^T A_B^{-1} A_N.$$

The basis remains optimal if:

$$\bar{c}_N \leq 0.$$

Thus, allowable changes in c must preserve the sign of reduced costs.

7.5 Allowable Range for Objective Coefficients

If we change only one coefficient c_k, the basis remains optimal as long as:

$$\bar{c}_j \leq 0 \quad \forall j \in N.$$

This gives upper and lower bounds for the allowable change.

Interpretation:
The objective direction can rotate slightly without changing the optimal vertex.

7.6 Change in Constraint Matrix A

If A changes, then both feasibility and reduced costs change:

$$x_B = A_B^{-1} b$$

$$\bar{c}_N = c_N - c_B^T A_B^{-1} A_N.$$

Small changes may preserve optimality, but structural changes may require re-optimization.

7.7 Parametric Linear Programming

In parametric LP, coefficients depend on parameter λ:

$$b(\lambda) = b + \lambda d.$$

The solution changes piecewise linearly in λ.

Important Result:
The optimal basis changes only at finitely many critical parameter values.

7.8 Multiple Optimal Solutions

If some reduced cost satisfies:

$$\bar{c}_j = 0$$

for $j \in N$, then there exists an alternate optimal solution.

Geometric Meaning:
The objective hyperplane is parallel to an edge of the feasible region.

7.9 Sensitivity Summary

- Changes in b affect feasibility.
- Changes in c affect reduced costs.
- Dual variables measure marginal value of resources.
- Optimal basis remains valid within allowable ranges.
- Optimal value changes linearly within stability region.

Deep Structural Insight:
Sensitivity analysis studies the local geometry of the polyhedron around the optimal vertex.

8 Advanced Structural Results in Linear Programming

This section develops deeper structural and theoretical aspects of Linear Programming, connecting convex analysis, polyhedral theory, and modern optimization.

8.1 Polyhedral Cones

8.1.1 Definition

A set $K \subseteq \mathbb{R}^n$ is a **cone** if:

$$x \in K \Rightarrow \lambda x \in K \quad \forall \lambda \geq 0.$$

A **polyhedral cone** is:

$$K = \{x : Ax \leq 0\}.$$

Important Example: The feasible directions at a BFS form a polyhedral cone.

8.2 Extreme Rays

A nonzero vector $d \in K$ is an **extreme ray** if:

$$d = d_1 + d_2, \quad d_1, d_2 \in K \Rightarrow d_1, d_2 \text{ are positive multiples of } d.$$

Connection to LP:
If d is an extreme ray along which

$$c^T d > 0,$$

then the LP is unbounded.

8.3 Minkowski–Weyl Theorem (Full Structure)

Theorem:
Every polyhedron P can be written as:

$$P = \text{conv}(V) + \text{cone}(R),$$

where:

- V = set of extreme points,
- R = set of extreme rays.

Special Case:
If P is bounded:

$$P = \text{conv}(V).$$

This theorem gives a complete structural description of polyhedra.

8.4 The Dual Cone

For a cone K, its dual cone is:

$$K^* = \{y : y^T x \geq 0 \ \forall x \in K\}.$$

Importance:
Dual feasibility conditions arise from dual cones.
Farkas' Lemma can be interpreted as a statement about dual cones.

8.5 Interior Point Methods

Unlike simplex, which moves along vertices, interior point methods move through the interior.

8.5.1 Barrier Method

Solve:

$$\max c^T x - \mu \sum \ln x_i$$

subject to $Ax = b$.

As $\mu \to 0$, solutions approach the boundary and converge to the optimal solution.

Key Insight:
Interior point methods are polynomial-time algorithms.

8.6 Complexity of Linear Programming

8.6.1 Combinatorial Complexity

Simplex may require exponential time in worst case.

8.6.2 Polynomial Time Result

Khachiyan (1979) proved LP is solvable in polynomial time using the ellipsoid method.

Karmarkar (1984) introduced a practical polynomial-time interior method.

Thus:

$$\text{LP is in } \mathbf{P}.$$

8.7 Total Unimodularity

A matrix A is **totally unimodular** if:

$$\det(B) \in \{-1, 0, 1\}$$

for every square submatrix B.

Important Theorem:
If A is totally unimodular and b is integral, then every BFS is integral.

Application:
This explains why network flow and assignment problems have integer solutions.

8.8 Sensitivity and Polyhedral Geometry

Optimal solution lies at:

- A vertex,
- Where objective hyperplane supports P.

Local geometry determines:

- Stability,
- Multiple optima,
- Degeneracy.

8.9 Connection with Convex Analysis

Linear Programming is a special case of convex optimization:

$$\min f(x) \quad \text{subject to } x \in C,$$

where:

- $f(x) = c^T x$ (linear),
- C is polyhedral convex.

Duality generalizes to:

- Lagrangian duality,

- Fenchel duality,
- KKT conditions.

LP is the simplest but structurally complete convex program.

Ultimate Structural Summary

Linear Programming is the study of:

Linear functionals over polyhedral convex sets.

Its structure is governed by:

- Linear independence,
- Convex geometry,
- Extreme points and rays,
- Supporting hyperplanes,
- Dual cones,
- Finite combinatorics of bases,
- Polynomial-time interior algorithms.

Grand Perspective:
LP sits at the intersection of:

- Linear Algebra,
- Convex Geometry,
- Optimization,
- Computational Complexity,
- Economic Equilibrium Theory.